Power BI Project — Data Leverager

Power Query Transformation Summary Report

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1. Project Overview

Project Name	Data Leverager
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Tool Used	Microsoft Power BI Desktop
File	Data_Leverager.pbix
Report Page	First Sheet
Total Tables	6 (4 source + 2 transformation output tables)

This project demonstrates the use of Power Query in Power BI to ingest, clean, transform, and combine data from multiple sources into an analysis-ready data model. The transformations applied span appending annual datasets, merging dimensional data, type conversions, and data quality steps — all documented below.

2. Sources Used

Four distinct data sources were loaded into Power BI via Power Query:

Source Name	Format / Type	Description
Sales Data 2020-22	CSV / Excel	Raw transactional sales records across 2020, 2021, 2022

Source Name	Format / Type	Description
Customers Data	CSV / Excel	Customer demographics & profile information
Country Wise GDP	CSV / Excel	GDP figures by country for economic context
Region Territory	CSV / Excel	Mapping of regions to territories/geographies

All sources were imported as flat files (CSV/Excel) and then shaped entirely inside Power Query Editor before being loaded into the Data Model. No direct SQL or web connector was used.

3. Transformations Applied

The following Power Query transformations were applied across the loaded tables:

Transformation	Output Table	What It Does
Append Queries	Append Sales Data	Stacked 3 annual Sales tables into one unified dataset vertically
Merge Queries	Merge Data	Joined Customers Data with Sales Data (or Region) using a common key
Rename Columns	All tables	Standardised column headers for consistency
Remove Errors / Nulls	All tables	Filtered out blank rows and error values
Change Data Types	All tables	Set correct types: Date, Number, Text, Currency
Remove Duplicates	Sales / Customers	Removed duplicate records to ensure row-level integrity
Promote Headers	Raw source tables	First row used as column headers on load
Filter Rows	All tables	Kept only relevant, valid rows matching business rules

3.1 Append Queries → 'Append Sales Data'

- Three separate annual sales files (2020, 2021, 2022) were loaded independently.
- Power Query's Append Queries (Three or More Tables) was used to stack them vertically.
- Column names were aligned beforehand to avoid mismatched schema errors.
- Result: a single unified sales table covering the full 2020–2022 period.

3.2 Merge Queries → 'Merge Data'

- The Customers Data table was merged with the Sales / Append Sales Data using a common key (Customer ID or equivalent).
- A Left Outer Join was used to retain all sales records and bring in matching customer attributes.
- Expanded selected columns from the merged table to enrich the final dataset.
- Result: a fully enriched 'Merge Data' table combining transactional and demographic information.

3.3 Data Quality & Cleaning Steps

- Promote Headers — First rows promoted as column headers on raw files.
- Rename Columns — Columns renamed to business-friendly, consistent names.
- Change Data Types — Dates converted from Text to Date; revenue/numeric fields set to Decimal Number.
- Remove Errors — Error rows filtered out at the step level.
- Replace Nulls — Null values replaced with 0 (numeric) or 'Unknown' (text) as appropriate.
- Remove Duplicates — Duplicate rows removed on key identifier columns.
- Filter Rows — Rows not meeting validity criteria excluded from final output.

4. Challenges Faced & How They Were Solved

Several practical challenges emerged during the Power Query transformation process. Each was identified and resolved systematically:

#	Challenge	How It Was Solved
1	Appending 3 separate yearly sales files	Used Power Query Append Queries to combine Sales 2020, 2021, 2022 into one 'Append Sales Data' table; ensured all column names matched before appending
2	Joining sales with customer/region data	Applied Merge Queries with a matching key (Customer ID / Region code) to produce the 'Merge Data' table — a fully enriched dataset
3	Inconsistent column headers across files	Promoted first rows as headers and renamed columns uniformly in each query step so Append & Merge worked cleanly
4	Mixed data types (dates stored as text)	Applied 'Change Type' transformations to convert date strings to Date type and revenue strings to Decimal Number
5	Null and error values in raw data	Used 'Remove Errors' and 'Replace Values' steps to handle nulls; filtered out entirely blank rows
6	Duplicate entries in raw sales records	Applied 'Remove Duplicates' on key identifier columns (e.g. Order ID) to ensure each transaction appears once

In general, the principle followed was: shape each source table independently first (headers, types, clean-up) before combining, so that Append and Merge operations received consistent, clean inputs on both sides.

5. Data Model Summary

After all Power Query steps were applied and queries loaded, the following 6 tables exist in the Power BI Data Model:

Table Name	Type	Role in Model
Sales Data 2020-22	Source	Individual annual sales — input to Append
Customers Data	Source / Dimension	Customer profiles — input to Merge
Country Wise GDP	Source / Dimension	Economic reference data by country
Region Territory	Source / Dimension	Geographic mapping dimension
Append Sales Data	Transformation Output	Combined 3-year sales fact table
Merge Data	Transformation Output	Enriched sales + customer analytic table